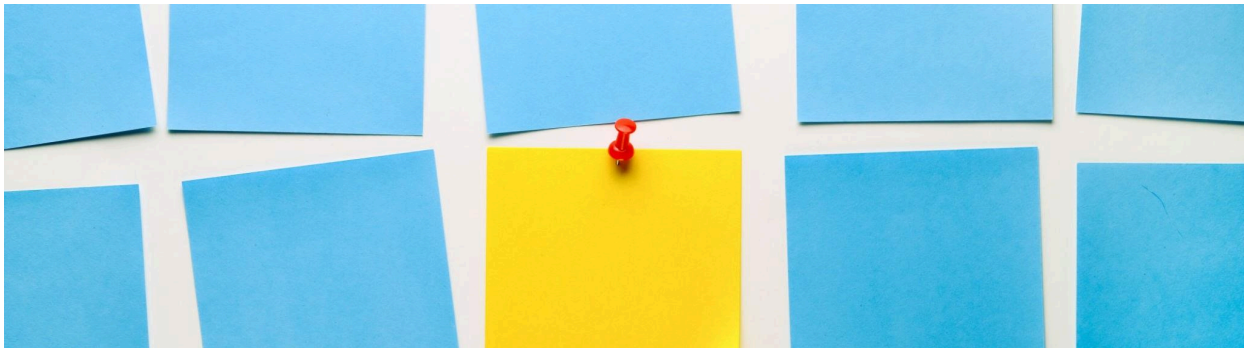

Project Chimera

Status In progress

Timing Apr 12, 2025 8:00 AM to Apr 13, 2025 1:00 AM

Overview

Project Chimera is a special initiative focused on bio-inspired technological solutions. The challenge asks participants to design a novel technology or system that mimics the principles of a natural organism or ecosystem to solve a modern problem. This aims to foster creative interdisciplinary thinking, blending biology and technology. Teams will develop a conceptual design and a basic interactive

prototype, presented alongside a detailed explanatory document.

Objectives

- Provide an overview of the vision, mission, and values: The vision is to encourage innovative problem-solving by looking to the natural world for inspiration. The mission is to challenge students to think across disciplines and develop ecologically sound technological concepts. Values include biomimicry, sustainability, innovation, and learning.
- Indicate key measurable metrics: Success will be measured by the novelty of the bio-inspired concept, the feasibility of the proposed solution, and the clarity of the presentation.
- Project objectives: Participants will ideate, design, and prototype a technology based on biomimicry principles. They will also create a detailed document outlining their concept and its biological inspiration.
- Define key success metrics: Projects will be evaluated based on Clarity of Biological Inspiration, Innovation of the Solution, Prototype Functionality , and Quality of the Explanatory Document.

Strategy

Approach

The challenge will begin with a brief introductory talk on biomimicry. Teams will then have the development period to research biological systems and design their solutions. Mentors with backgrounds in biology and engineering will be available for guidance. Submissions will include a prototype and a written document detailing the biological inspiration, design process, and potential impact.

Target audience

- High school students with an interest in both STEM and natural sciences. Teams of 3-4 members are required.